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The spatial planning of public electric vehicle charging infrastructure in a high-density city using a contextualised location-allocation model

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ABSTRACT

The optimal deployment of public charging infrastructure is critical to the popularisation of electric vehicles (EVs) in high-density cities. Existing studies on public EV charging facilities have rarely integrated government policy and spatial constraints into their optimisation algorithms. To address this research gap, we proposed a contextualised EV charger optimisation model that incorporates carefully derived supply-and-demand constraints and tested it in the case of Hong Kong. From the supply side, we studied the latest planning guidelines and conducted a spatial analysis of potential charging sites. From the demand side, we conducted a questionnaire survey with local residents, estimated their EV purchase intention using a generalised ordered probit model, and then projected the usage demand for public chargers. These supply-and-demand constraints were subsequently incorporated into a location-allocation model to minimise both charging demand shortfall and travel time to charging facilities. We also conducted sensitivity analyses with varying budget, charging demand and facility service radius. Based on our results, we made several key recommendations regarding the spatial planning of public EV charging facilities in our high-density context: (1) the existing charging network should be substantially expanded to meet the projected demand; (2) the charging network should be expanded beyond the central business district and the urban core into other urban neighbourhoods and suburbs; and (3) installing more chargers at existing charging stations is more economical than building new stations. Our research provides an important reference for the spatial planning and deployment strategies for public EV charging infrastructure in high-density cities.

1. Introduction

Electric vehicles (EVs) have received worldwide attention as a sustainable alternative to internal-combustion-engine vehicles (ICEVs) by mitigating roadside carbon dioxide emissions, improving energy efficiency, reducing oil dependency and enhancing passenger experience (Jia et al., 2019; Lee & Madanat, 2017; Napoli et al., 2020; Plötz et al., 2014). Global EV numbers reached 10 million at the end of 2020, indicating a 41% increase from the previous year despite the pandemic-related contraction of the vehicle market more generally (International Energy Agency, 2021). Meanwhile, several countries – including Norway, the UK, Germany and Singapore – have pledged to ban cars running on fossil fuels between 2025 and 2040 (Bennett & Vijaygopal, 2018; Dugdale, 2018; Tan,

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Studies have shown that EV adoption depends considerably upon the continuous expansion of charging networks, with adequate access to charging facilities necessary to alleviate range anxiety (Abotalebi et al., 2018; Dong et al., 2019; He et al., 2016; Kong et al., 2017; Napoli et al., 2020). Providing public charging infrastructure is especially crucial for high-density cities (He et al., 2016), where home charging is often unfeasible due to the scarcity of single-family housing units and the lack of individual parking space. Patt et al. (2019) predicted that automobile owners who own a private parking space are 50% more likely to adopt EV than those who share parking lots. Importantly, the high-density living that characterises the prevalent urban landscape in Asia (Liu et al., 2016) implies strong reliance on public charging networks – along with reduced ease of home charging – rendering Western insights less applicable.

Nonetheless, in recent years, there has been increasing attention on EV charging networks in Asian markets. For example, Liu (2012) considered the impact of fast-charging points and battery-swap stations on the power grid in Beijing, China. Working also on Beijing, Cai et al. (2014) gauged improvements in the electrification rate and the reduction of emissions derived from optimally locating public charging stations. Elsewhere, various studies have estimated EV charging demand using taxi trip data. For instance, Liu et al. (2016) devised an optimal network that minimised the discomfort of EV drivers by capturing the station seeking and queueing experience using taxi trajectory data, and Zhu et al. (2016) proposed a model minimising travel costs between trip destinations and assigned charging stations using Beijing as a case study. Han et al. (2016) planned a cost-minimising model for optimal taxi charging locations in metropolitan Daejeon, South Korea, with additional considerations of passenger presence and battery performance data, and a study in the Singaporean context demonstrated how charging infrastructure optimisation can be integrated into public housing car parks by simulating the electrification of taxi fleets (Wang et al., 2019). Meanwhile, rather than assigning facilities over a single time period, Chung and Kwon (2014) highlighted the effect of investment timetables on infrastructure expansion in the context of multi-period planning of EV charging facilities in South Korea.

However, consideration of the extant literature reveals several research gaps. First, most existing studies have been dedicated to explaining problem formulation and computational procedures (e.g. Chen et al., 2017; Chung & Kwon, 2015; J. He et al., 2018). Although empirical case studies have been presented, they have often been treated as examples to validate the performance metrics generated by the optimisation models. Thus, spatial analysis of model results remains underdeveloped, and to the best of our knowledge, few studies have further interpreted optimal deployment plots and indicated facility planning implications. Second, in terms of the spatial planning perspective, limited effort has been dedicated to contextualising the facility location problem. Given charging facility utilisation and urban forms differ across geographies, contextualising charging supply-and-demand constraints is essential to emulating the simulated localities. However, neither institutional nor planning guidelines have been incorporated into the establishment of optimisation models. Instead, the literature often disassociates parameters for localising deployment scenarios – such as EV penetration rate (Frade et al., 2011; Ghamami et al., 2016; Vazifeh et al., 2019; Wang et al., 2019), station coverage radius (Cai et al., 2014; Liu, 2012; Tao et al., 2018) and station capacity (Dong et al., 2014; Frade et al., 2011; Liu, 2012) – from actual planning guidelines or policy documents. Therefore, the optimised solutions might deviate from the facility deployment plan of the studied geographies, undermining the practical value of these solutions. Third, most existing studies have planned for optimal facility location based on the charging demand status quo (e.g. Dong et al., 2019; Jia et al., 2012), with only a few studies estimating future charging demand (Frade et al., 2011; He et al., 2016; Namdeo et al., 2014). Given the planning and deployment time for infrastructure provision, optimisation models considering only present data may not inherently resemble the future demand of EV charging facilities.

This study attempts to address these research gaps by developing and estimating a location-allocation (LA) model that carefully incorporates supply-and-demand estimation of EV charging. Supply constraints are derived from planning guidelines and existing practices in our study area to reflect planning elements for more pragmatic value of this empirical study, while charging demand of EV is estimated based on a stated preference survey. The rest of the paper is structured as follows: Section 2 reviews the literature on EV charging facility LA problems and introduces the theoretical grounding for the methodologies employed; Section 3 describes this paper's data collection process and research methodology; Section 4 reports findings from our EV demand projection and introduces our LA optimisation solution; Section 5 presents three sensitivity tests conducted on the LA model; finally, Section 6 discusses the policy implications of our findings.

2. Literature review

2.1. Charging facility optimisation studies

Broadly speaking, optimising EV charging facilities is a facility location problem, which requires selecting the optimal sites for facilities in a given region to realise various allocation objectives (Weber, 1929; Bai et al., 2019). In terms of analytical frameworks, extensive studies have applied LA models to identify optimal locations for EV chargers (e.g. Andrenacci et al., 2016; Baouche et al., 2014; Csiszára et al., 2019; Frade et al., 2011; He et al., 2016; Kong et al., 2017; Liu et al., 2019; Zhang et al., 2019).

Electric vehicle charging facility LA problems can be divided into node-based and flow-based problems, which are chosen depending on the form of demand definition. Node-based models, assuming facility demand is stored at discrete points, can be further divided into p-median models and set coverage models. The p-median problem was first introduced by Hakimi (1964) and deploys p number of facilities to minimise total travel cost. The set coverage problem locates the minimum number of facilities required to cover all demand points based on a defined facility service radius (Toregas et al., 1971). Studies expressing facility demand in terms of traffic flow along the network adopt the flow-capturing model proposed by Hodgson (1990) to deploy facilities at locations where the maximum traffic flow could be intercepted.

Research efforts have consistently aimed to incorporate specific decision-making rules into the LA model to yield more realistic siting recommendations. For example, Ghamami et al. (2016), Jia et al. (2012) and Perera et al. (2020) have all imposed minimisation objectives on both consumer travel cost and facility provision cost to integrate supply-side considerations. Elsewhere, Jia et al. (2019) devised a two-step optimisation model to minimise total travel time from demand clusters to charging stations, and Vazifeh et al. (2019) expanded the set covering objective to minimise charging station volume, excess driving distance from trip end-points to stations and energy overheads. Meanwhile, extending the flow refuelling model, Erdoğan et al. (2022) tailored a corridor-based problem and demonstrated that it outperformed the traditional objectives in a case considering candidate locations along and outside the corridor.

The LA analysis in the studies mentioned above were typically solved under a series of constraints. The following subsections review the imposition of these constraints in literature, which can be categorised into supply- and demand-side constraints.

2.2. Charging supply analysis

Studies have contemplated a range of locations as potential sites for deploying EV charging facilities. Depending on the optimisation target, supply locations can be classified as continuous location, network location or discrete location (Zhang et al., 2019), with most studies considering discrete supply location, where discrete points are reserved as candidate sites for EV charger deployment. Two frequently adopted candidate locations are parking lots (De Gennaro et al., 2015; Dong et al., 2019; Frade et al., 2011; Tao et al., 2018) and petrol stations (Cai et al., 2014; Liu et al., 2019), with some studies combining the two (Liu, 2012; Wang et al., 2019). Furthermore, several studies have optimised facility distribution along highway corridors, often adopting a continuous or network location approach to deploying charging facilities at vertices and the edges of road networks (e.g. J. He et al., 2018; Napoli et al., 2020). All of these potential locations have been incorporated into LA models as supply-side constraints on selecting optimal sites for EV charging facility deployment.

These constraints specify the technical attributes of chargers and the rules of station deployment. At the network level, allocation scripts manipulate the service radius of charging stations and the types of chargers installed. For example, Liu (2012) derived a 2-km service radius for EV charging stations based on petrol station distribution to prevent supply-coverage redundancy; meanwhile, other studies have defined the service radius according to governmental planning guidelines (He et al., 2016; Jia et al., 2019). Various studies have facilitated discussion by distinguishing the types of charging facilities to be deployed (Chen et al., 2017; De Gennaro et al., 2015; Wolbertus et al., 2021) and the varying rates of home charging availability (Cai et al., 2014; Dong et al., 2014; Liu, 2012; Pan et al., 2020). At the station level, charging station capacity has been controlled to ensure reasonably even supply distribution. For example, Frade et al. (2011) confined the deployment scale from two to ten outlets per site to avoid immobilising the power grid.

2.3. Charging demand analysis

Representations of facility demand pertain closely to the choice of facility location problems, with most optimisation objectives either fully or partially connected to facility demand. Accordingly, research on the EV charging infrastructure LA problem needs to first properly define and estimate EV charging demand. Electric vehicle charging demand is often gauged using three information sources: vehicle user attributes, trip origin–destination (O-D) diary and vehicle flow GPS data (Pagany et al., 2018). Data sources are converted into flow-based or node-based representations to enable estimation of charging demand across space (Bräunl et al., 2020; Shen et al., 2019).

Although flow-based demand construction has primarily utilised continuous trajectory data from vehicle trip chains (Bräunl et al., 2020; Cai et al., 2014; De Gennaro et al., 2015; Liu et al., 2019; Tao et al., 2018), in some cases, O-D data has been utilised after route assignment (Dong et al., 2014; Erdoğan et al. 2022; Ghamami et al., 2016; Jia et al., 2019; Namdeo et al., 2014; Vazifeh et al., 2019; Xi et al., 2013). Researchers have then hypothetically electrified part or all of the travel-path data to estimate EV charging needs (Andrenacci et al., 2016; Bräunl et al., 2020; Cai et al., 2014; De Gennaro et al., 2015; Ghamami et al., 2016). Nevertheless, there are limitations to using network flow information as the charging demand proxy. First, projecting charging demand based on extant vehicle flow assumes convention vehicle flow patterns replicate those of EVs (Bräunl et al., 2020; De Gennaro et al., 2015), a strong assumption that has yet to be empirically validated. Second, complex road networks within densely developed areas generate unordered flow, which complicates predictions of traffic flow (Bai et al., 2019).

Another approach to estimate EV charging demand is node-based demand estimation using demographic data. Given charging activities often occur at trip destinations where a full charge is feasible – for example, the home or workplace – node-based proxies for EV charging demand are considered more suitable. Such demand estimation approaches aggregate zonal demand (Bai et al., 2019; Frade et al., 2011; He et al., 2016; Pan et al., 2020; Perera et al., 2020). However, given the limited number of EV users in most cases studied, city-wide surveys interrogating the target group's spatial distribution remain scant. Previous studies have often projected EV demand based on personal or household characteristics using statistical models such as regression analyses (Campbell et al., 2012; Namdeo et al., 2014).

3. Methods: Location-allocation model

An overview of the methodology adopted in this study is illustrated in Fig. 1. The workflow of EV charging facility allocation is divided into supply-side and demand-side analyses: while the supply-side analysis was carried out to articulate the preconditions of charging infrastructure provision, demand-side analysis was carried out to project the distribution and intensity of EV charging needs. Output from the supply and demand analyses, and a series of constraints derived from ancillary analyses were then used to formulate

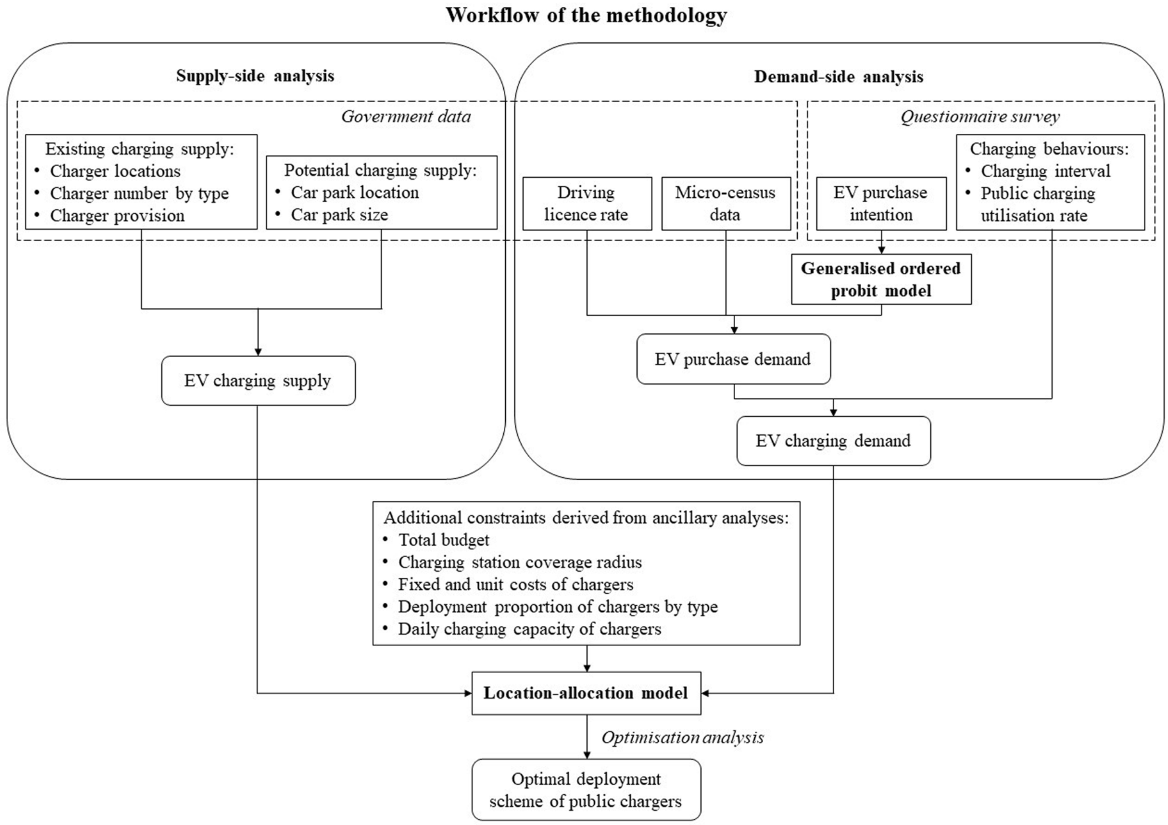

Fig. 1. Workflow of proposed methodology.

and solve an optimisation problem in our LA model in order to estimate the optimal deployment scheme of public EV charging facilities.

This study considers Hong Kong, a high-density city with a population of 7.5 million people and a built-up area of 270 square kilometres that represents less than 25% of the territorial land area. The urban centres of the city are located along the harbourfront areas of Hong Kong Island and Kowloon. Outside the urban areas, new towns and new development areas constitute the suburban areas of the city, which has lower job accessibility and longer commutes because of the strong agglomeration economy in the urban core (He et al., 2020a; He et al., 2020b). In the following sections, we conduct supply and demand analyses at the neighbourhood level, using large street block groups (LSBGs) – the finest spatial analysis unit defined by the Hong Kong Census and Statistics Department and used in population censuses – to approximate neighbourhoods, ultimately identifying a total of 1,622 LSBGs.

Terms used to describe charging facility provision in this paper are defined as follows. A charger is an individual device supplying electric energy to EVs, which is sometimes referred to as charging pile or electric vehicle supply equipment (EVSE) in the literature. A charging station is a venue where one or more chargers are installed, typically located in a car park in our study area. Charging facilities and charging infrastructure can refer to either chargers or charging stations. Multiple charging stations form a charging network.

3.1. Supply-side analysis: Existing and potential electric vehicle charging stations

Hong Kong's existing EV charging network is spread across car parks in an array of building types registered within seven building categories pertaining to five land-use classes; Appendix 1 summarises this information. As of October 2019, there were 2,503[1] public EV chargers in Hong Kong (Hong Kong Environmental Protection Department, 2019). Depending on charging power, government definitions classify EV chargers as standard (N = 923), medium (N = 1003) or quick (N = 577) chargers. Standard chargers are charging appliances that connect to a household 13A socket. Medium chargers provide electric power at a level not higher than 20 kW. Quick chargers supply electricity at levels above 20 kW. In terms of US (and European) specifications, standard chargers in Hong Kong correspond to level 1 (slow) chargers, medium chargers correspond to certain level 2 (fast) chargers, and quick chargers correspond to both higher-powered level 2 chargers and level 3 (rapid) DC chargers. Charging point particulars were retrieved from official records

[1] Excluding three chargers exclusively for the use of Tesla Roadsters.

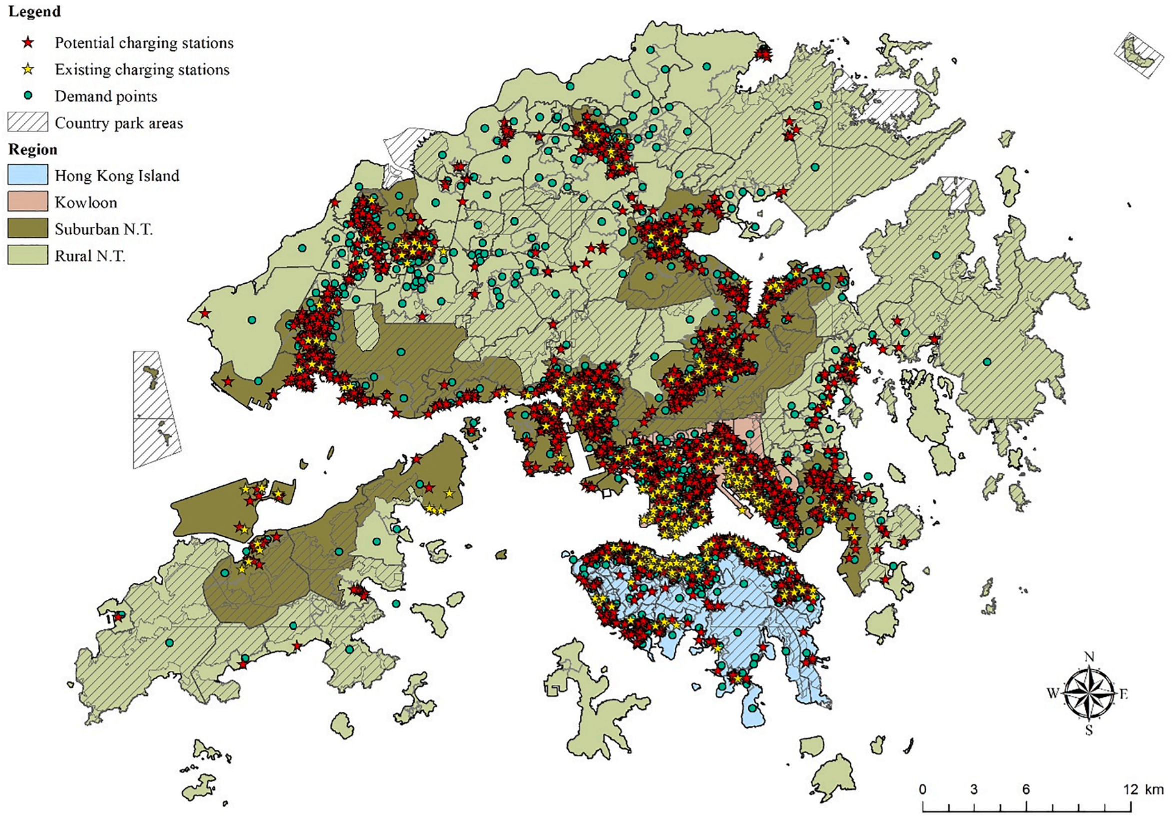

Fig. 2. Distribution of charging supply and demand in Hong Kong.

for each station; these particulars included physical address, the number of connectors at the station and the charging speed of the connectors. The provision of publicly accessible charging facilities in Hong Kong is financed by either the government or the private sector. As of October 2019, 34.2% of public chargers were government-provided, with the other 65.8% built by non-government entities. Each government charging point supplied an average of 16.48 charging outlets; at non-government charging stations, the average was 5.77.

Moving beyond the current charging network, we have identified public car parks as potential sites for charging facility deployment because the whole of Hong Kong's existing public charging network corresponds to parking lots. Although the literature indicates that places other than car parks could be included as potential charging sites – with petrol stations a commonly employed location type in certain contexts (Cai et al., 2014; Liu et al., 2019) – as of March 2021, no EV chargers have been installed at petrol stations in Hong Kong (Hong Kong Environment Bureau, 2021). Therefore, for feasibility and practicality reasons, only public car parks have been considered viable charging supply sites by our study.

A dataset from the Planning Department of Hong Kong was used to generate the pool of potential sites. Then, the spatial attributes of the recorded car parks were augmented by parking space information and charging point particulars, producing an operational dataset comprising 2,295 car parks, of which 332 featured EV charging facilities. Fig. 2 presents the distribution of existing and potential charging sites. Although car park locations generally overlapped with Hong Kong's urban areas, EV charging stations were concentrated along the harbourfronts of Hong Kong Island and Kowloon Peninsula. In contrast, charger supply in the New Territories was scattered across satellite communities in suburban areas, with few existing or potential charging spots located in the relatively vast rural region.

3.2. *Demand-side analysis*

Distribution of EV charging facilities is principally determined by EV charging demand distribution (Liu et al., 2016). Therefore, to choose the most efficient locations for charging services, it was first necessary to understand the spatial distribution of EV charging demand. The demand-side analysis included two steps. First, we predicted EV purchase demand for each neighbourhood in the study area based on the number of EVs. Second, we estimated each neighbourhood's charging needs based on the prospective EV fleet.

3.2.1. *Data collection: Questionnaire survey*

For EV purchase demand, we measured a person's intention to buy an EV in the next 5 years using a 7-point Likert scale where 1

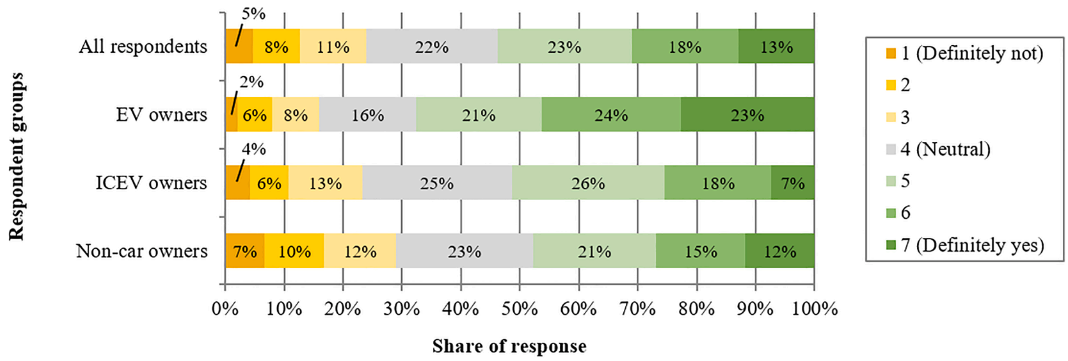

Fig. 3. EV purchase intention within the next 5 years according to vehicle ownership group.

corresponded to 'definitely not' and 7 corresponded to 'definitely'. An online survey was conducted in Hong Kong between March and September 2019. Survey respondents were recruited from different sources, including a survey company's online survey panel, EV driver associations, social media communities and online forums. Excluding the surveys conducted by the survey panel, responses were provided on a voluntary basis. Citizens without a driving licence were excluded from the survey to avoid distorting affinity for EV and ensure sufficient understanding of the research topic (Carley et al., 2013; Schuitema et al., 2013; Zhang et al., 2011). Additionally, given the low population share of EV drivers, we oversampled that group to attain a proportion of 25%, emulating a recent survey in Denmark (Thøgersen & Ebsen, 2019). This approach is often used to reveal the behaviour of an under-represented but interesting population stratum (e.g. Klöckner et al., 2013; Langbroek et al., 2017). Our LA model derives certain charger usage parameters – including charging interval (once every 4 days on average) and public charger usage ratio (50%) – from the responses of EV owners to our questionnaire survey. Accordingly, we oversampled the EV owner group to guarantee a reliable inference.

Our online survey received a total of 982 valid responses. Compared to the average adult population (residents aged 18 or above) in Hong Kong, our respondents were more likely to be economically advantaged, well-educated, young and male, features that conform to common observations of the traits of private car drivers and EV adopters (e.g. Axsen et al., 2016; Plötz et al., 2014; Tal & Nicolas, 2013).

Appendix 2 presents a summary of the personal and household characteristics of survey respondents. The sample was divided into EV owners (N = 238), ICEV owners (N = 327), and those without cars (N = 417). The EV-owner group featured the highest proportion of males, with the gender split more even among ICEV owners. Electric vehicle owners tended to have a higher level of education, featuring the highest proportion of postgraduate degree holders and the lowest proportion of those with an education level below undergraduate. Regarding employment status, EV owners were twice as likely to be employers or self-employed. Accordingly, their economic status also exceeded the two other groups, with half of the EV drivers earning over HK$40,000 per month and 75% living in private residential units. The proportion of renters among EV owners was the smallest of the three groups. Fig. 3 depicts 5-year EV purchase intention, the dependent variable used for demand modelling; here, as expected, existing EV owners expressed strong EV adoption inertia. For the whole sample, more than half of respondents (54%) were inclined to buy an EV in the near future, and 13% indicated a definite propensity.

3.2.2. Estimating purchase demand for electric vehicles

For EV purchase demand, the dependent variable was a person's intention to buy an EV in the next 5 years, measured using a 7-point Likert scale. We estimated factors leading a respondent to 'definitely' intend to buy an EV (i.e. those who indicated the highest possible response on the Likert scale: $y = 7$), with the derived explanatory variables including individual characteristics (i.e. age, gender, education, marital status, employment status and income), household characteristics (i.e. housing type, tenure type, household size and number of children under the age of 15), residential district and workplace district.

A common approach to managing ordered alternatives is an ordered logit model. Note that the ordered logit model is essentially an ordinal model. The strongest assumption of ordinal models is the proportional odds assumption (also known as the parallel lines assumption), which indicates that all explanatory variable coefficients are identical across all outcome categories. If this assumption is violated, a better solution is, arguably, the generalised logistic regression model (or partial proportional odds model) (Peterson & Harrell, 1990). This study tested the proportionality of odds assumption, observing that the assumption did not hold for our data. To address this issue, one solution is generalised ordered logistic (gologit) model, which allows some of the coefficients to be the same for all outcome categories, while others can differ. In other words, it can estimate partial proportional odds models (Williams, 2016).

Following William's (2006) notation, the gologit model can be written as:

$$prob(Y_i > j) = g(X\beta_j) = \frac{\exp(\alpha_j + X_i\beta_j)}{1 + [\exp(\alpha_j + X_i\beta_j)]}, \quad j = 1, 2, \cdots, J - 1 \tag{1}$$

where α is the constant term, X refers to the explanatory variables, β refers to the coefficients of the explanatory variables, J is the number of categories of the dependent variable, which is 7 in our case. The probabilities Y will take can be determined by the following:

$$prob(Y_i = 1) = 1 - g(X_i\beta_1)$$

$$prob(Y_i = j) = g(X_i\beta_{j-1}) - g(X_i\beta_j), \quad j = 2, \cdots, J-1$$

$$prob(Y_i = J) = g(X_i\beta_{J-1})$$

For our study, we used a specific model type under the generalised logistic model – the generalised ordered probit model. Our model was estimated in STATA using a user written package 'gologit2′ (Williams, 2006).

After estimating the probability of an individual buying an EV using the generalised ordered probit model with our survey data, we applied the estimated coefficients from the generalised ordered probit model to the population in the community to predict aggregate demand using the sample enumeration procedure (Ben-Akiva & Lerman, 1985; Koppelman, 1974). More specifically, we predicted the aggregate probability of a person of a community i intending to buy an EV,a_i, by averaging individual probabilities according to the following:

$$a_i = \frac{1}{N_i}\sum_{n=1}^{N_i} Prob(y = 7|\boldsymbol{w_{in}}) \tag{2}$$

where N_i indicates the number of qualified individuals (i.e. residents aged 18 or older) from community i (based on 2016 micro-census data[2] from the Hong Kong Census and Statistics Department), and $\boldsymbol{w_{in}}$ denotes the socio-demographic attributes of the n-th individual from community i (approximated by LSBG).

Based on the estimated community-level probability a_i that an individual will purchase an EV, a community's aggregate EV purchase demand can be obtained by multiplying the aggregate probability (a_i) by that community's total population of interest (in this case, residents over 18 years of age with a driving licence). Given Hong Kong's census features no information about driving licences across geographical units, we used a territorial average, assuming that all communities would feature roughly the same share of licensed drivers (30.8%[3]).

3.2.3. Estimating charging demand from electric vehicles

After estimating EV purchase demand, we projected daily demand for public charger use, using the following formula for public charging facility usage intensity (D^d):

$$D^d = D^*P^{d*}P_{pub} \tag{3}$$

where D indicates purchase demand for each neighbourhood (see **Section 3.2.1**), P^d represents the percentage of EVs charging on a single day, assuming the same rate for all neighbourhoods, and P_{pub} denotes the percentage of EVs using public chargers, assuming the same rate for all neighbourhoods.

We estimated P^d and P_{pub} according to the charging behaviour of EV owners who participated in our questionnaire survey. The questionnaire inquired about their regular charging interval and their primary charging location. Based on the survey results, EV drivers on average charge their EVs at a 4-day interval, and about 52.5% of them (N = 125) utilised the public charging network as primary charging solution. Therefore, we assigned a value of 0.25 for P^d and a value of 0.5 for P_{pub} for Eq. (3).

3.3. Location-allocation model

The LA model was developed according to the constraints identified by our supply and demand analyses. To allocate charging stations, the model connected EV charging demand projection with existing/potential supply locations in the form of O-D matching. To enable O-D matching, areal street block units capturing estimated neighbourhood-level charging demand were converted into a set of demand points using the centroids of the spatial units. Based on context, our model considered whether a given car park should be equipped with EV charging facilities, how many chargers should be installed (in addition to existing chargers), and how geographical demand points should be allocated to car parks with charging facilities. A demand point's charging demand could be split between multiple car parks.

Our model utilised the following sets and parameters:

Demand side (estimated by the demand estimation model)

I = set of demand points;

h_i = estimated number of EVs requiring charging each day at demand point i;

Supply side (estimated from the Government/public data and in-depth interview)

[2] Similar to micro-census (or micro-data) datasets from other countries/regions, this dataset comprised personal and household records alongside location information (e.g. street block, census tract, district).

[3] The ratio was calculated based on 2,319,906 full-licence holders as of September 2019 (Hong Kong Transport Department, 2019) – and the total population of 7,524,100 as of mid-2019 (Hong Kong Census and Statistics Department, 2020).

J = set of car parks (including both car parks with existing EV chargers and car parks with potential for charger installation);
$\bar{J}$ = set of car parks with existing chargers;
K = set of charger types (standard, medium and quick);
L = set of land-use types (i. residential buildings or complexes; ii. business or commercial centres and hotels; iii. government or institutional buildings; iv. comprehensive development areas; v. industrial centres; vi. transport hubs; vii. public amenities);
q_{jk} = number of existing type k chargers at car park j;
Q_j = maximum number of chargers at car park j;
μ_k = number of EVs that can be charged per day per type k charger;
$\overline{m}_k$ = maximum number of type k chargers to be installed;
m_{-k} = minimum number of type k chargers to be installed;
$\overline{p}_k$ = maximum proportion of type k chargers to be installed;
p_{-k} = minimum proportion of type k chargers to be installed;
D = a charging facility's coverage radius;
J_i = set of car parks located within the coverage radius of demand point i (i.e. $J_i = \{j \in J : d_{ij} \le D\}$);
J_l = set of car parks that are located in a type l land-use area;
f_j = fixed cost of establishing charging facilities at car park j (0 if car park j features existing charging facilities);
c_{jk} = unit cost of installing a type k charger at car park j;
B_l = budget for establishing new charging facilities for type l land use;
M = penalty for charging demand shortfall (per EV);
Transport network analysis
d_{ij} = driving distance between demand point i and car park j;
t_{ij} = driving time between demand point i and car park j;
Note: The values of supply- and demand-side parameters were estimated following the workflow of our methodology (see Fig. 1).

Our model optimised the following decision variables:

x_{ij} = number of EVs from demand point i allocated to car park j;

s_i = charging demand shortfall at demand point i;

$$y_j = \begin{cases} 1 & \text{if car park } j \text{ is equipped with EV charging facilities,} \\ 0 & \text{otherwise} \end{cases};\ \text{and}$$

z_{jk} = number of type k chargers to be installed at car park j.

The LA model was formulated as follows:

$$Min \quad M\sum_{i\in I} s_i + \sum_{i\in I, j\in J} t_{ij}x_{ij} \tag{1}$$

subject to

$$\sum_{j\in J_i} x_{ij} + s_i = h_i \quad \forall i \in I \tag{2}$$

$$\sum_{i\in I} x_{ij} \le \sum_{k\in k} \mu_k \left(q_{jk} + z_{jk}\right) \quad \forall j \in J \tag{3}$$

$$\sum_{k\in K} \left(q_{jk} + z_{jk}\right) \le Q_j y_j \quad \forall j \in J \tag{4}$$

$$\sum_{j\in J_l} \left(f_j y_j + \sum_{k\in K} c_{jk} z_{jk} \right) \le B_l \quad \forall l \in L \tag{5}$$

$$m_{-k} \le \sum_{j\in J} z_{jk} \le \overline{m}_k \quad \forall k \in K \tag{6}$$

$$p_{-k} \le \frac{\sum_{j\in J} z_{jk}}{\sum_{j\in J}\sum_{k' \in K} z_{ik'}} \le \overline{p}_k \quad \forall k \in K \tag{7}$$

$$y_j = 1 \quad \forall j \in \bar{J} \tag{8}$$

$$x_{ij} = 0 \quad \forall j \in J \backslash J_i \tag{9}$$

$$y_i \in \mathbb{B} \quad \forall j \in J \tag{10}$$

$$z_{jk} \in \mathbb{Z}^{+} \quad \forall j \in J, k \in K \tag{11}$$

$$x_{ij}, s_i \geq 0 \quad \forall i \in I, j \in J \tag{12}$$

The objective of (1) was to minimise both the penalty for EV charging demand shortfall and the time required to travel to charging facilities, weighted for demand. M was set larger than t_{ij} because the first priority was minimising EV charging demand shortfall. Next, the accessibility of charging facilities was optimised. However, to avoid computational convergence issues, the value of M should not be assigned an excessive value. Thus, our case study set the value of M to 30 (minutes); although greater than all$\{t_{ij} : j \in J_i\}_{\forall i \in I}$, it maintains a similar travel time scale to ensure efficient computation. The next 11 equations represent constraints on the LA model: (2) allocated the charging demand of a demand point to charging facilities within the coverage radius (if demand could not be fully satisfied by the charging facilities, charging demand shortfall was measured); (3) imposed the condition that the total charging demand allocated to each car park should not exceed the maximum number of EVs that could be charged each day at the car park; (4) imposed space constraints limiting the number of chargers at each car park; (5) imposed a financial budget constraint for establishing new charging facilities in areas of each location type; (6) and (7) set the ranges for the number and proportion of each charger type to be installed; (8) recorded car parks with existing charging facilities; (9) ensured that a demand point's demand could not be allocated to a car park beyond the coverage radius; (10–12) stated the nature of each decision variable.

Regarding problem size, although the formulation contained four types of decision variables ($x_{ij}, s_i, y_j,$ andz_{jk}), the maximum number of indices a decision variable type contained could more significantly impact problem size due to 'the curse of dimensionality' (Bellman, 1957). The maximum number of decision variables was$|I||J| + |I| + |J| + |J||K|$, some of which could be eliminated before solving the problems due to the coverage radius. Meanwhile, almost all constraints were associated with only one index in the enumeration domain; exceptions were those requiring definition of decision variable domains, namely, constraints (8–12). Thus, our formulation capitalised on the manageability of problem size even when the numbers of communities, charging locations and charger types increase, which is not the case for other classes of optimisation problems. The decision variables requiring most computational effort were binary and integer decisions (i.e. y_j, z_{jk}); this is due to the branch-and-cut approach's branching procedure. Thus, our formulation's problem size can be managed using many off-the-shelf optimisation solvers (e.g. CPLEX in our computation study).

Table 1 presents the parameters used for our study's baseline model. For demand, only respondents who indicated 7 for purchase intention were included, ensuring the model considered only the most likely purchasers. For charger-level supply capacity, we chose two popular EV models with mid-range battery capacities[4] to estimate the number of EVs to be charged each day by each charger type. Overnight charging (8 hours) was disabled on quick chargers because their short charging cycle meant that we did not expect overnight turnover despite the capacity enabled by their charging speed. To simulate contextual demand allocation and supply distribution, we imposed a series of station-level supply constraints. First, using the charging facility planning instructions of Beijing as a reference[5], service radii for charging stations (by region) were stipulated in terms of driving distance. Second, charging stations were kept to a maximum of eight chargers to avoid over-concentration of charging supply, a constraint based upon the average number of chargers at public charging stations in October 2019. Third, to reflect the Hong Kong government's public charger installation forecast, our baseline scenario distributes only medium and quick chargers (Hong Kong Environmental Protection Department, 2019; Hong Kong Productivity Council, 2018). Fourth, at least 739 medium chargers were designated for allocation, reflecting the prescribed deployment schedule for public chargers in government car parks up to March 2021 (Hong Kong Environmental Protection Department, 2020a). Fifth, a constraint on the deployment ratio of medium to quick chargers (52:48) was applied to non-government charging points to match the October 2019 ratio. Given the maximum number of standard chargers was set to 0, we modified constraint (7) to incorporate it into our model[6], producing the following:

$$0.515 \leq \frac{\sum_{j \in J} z_{jk} - 739}{\sum_{j \in J} \sum_{k' \in K} z_{ik'}} \leq 0.525 \quad (\text{where} \ \ k = \text{medium} \ \ \text{charger}) \tag{7}$$

Budgets for the two provision sources (government and private sector) were estimated separately before being combined for the model. Given the exact number of chargers to be distributed in government car parks – 739 medium chargers – was dictated, charger

[4] The two models were the Tesla Model 3 Standard Range+ (55kWh) and Nissan Leaf (40kWh). The Tesla vehicle requires 40 min to an hour at a DC quick charger to reach a complete charge from 0% battery power, 7 h at a medium charger and 16 h at a standard charger. The Nissan vehicle requires 8 h at a medium charger and 18 h at a standard charger. The converted daily service capacity for each type of charger conforms to estimates provided by charging-service providers.

[5] The *Action Plan for Promoting EVs in Beijing (2014–2017)* specified a service radius of 5 km for public charging facilities (The People's Government of Beijing Municipality, 2014). The coverage requirement was then weighted according to the population density of regions in Hong Kong to impose the service radii.

[6] We have conducted several sensitivity analyses; although some are reported in this paper's latter sections, some are not reported due to space constraints. In some of the additional sensitivity analyses, we allowed the number of standard chargers to be greater than 0; in such cases, constraint (7) rather than constraint (7′) was incorporated into the model.

Table 1
Parameters for the case study's baseline scenario.

Parameter	Value
Demand	
Purchase intention score (used to estimate EV charging demand)	7
Supply	
Charger-level constraints	
Number of EVs that can be charged by a standard charger per day	1
Number of EVs that can be charged by a medium charger per day	3
Number of EVs that can be charged by a quick charger per day	16
Coverage distance of charging stations in urban areas (driving distance)	1.6 km
Coverage distance of charging stations in suburban areas (driving distance)	2 km
Coverage distance of charging stations in rural areas (driving distance)	10 km
Maximum number of chargers at a charging station	8
Minimum numbers of standard/medium/quick chargers	0/739/0
Maximum numbers of standard/medium/quick chargers	100,000/100,000/100,000
Minimum proportion of standard/medium/quick chargers	0/51.5%/47.5%
Maximum proportion of standard/medium/quick chargers	0/52.5%/48.5%
Budget and cost estimates	
Total budget	HK$ 130,253,000
Fixed cost of establishing charging facilities at a car park	HK$ 43,500
Unit cost of a medium charger	HK$ 16,000
Unit cost of a fast charger	HK$ 136,000

costs and predicted fixed costs – based on chargers per station[7] – could be added to calculate the government's domain budget. Given the absence of any record or plan specifying the financial resources dedicated to private car park installations, spending on non-government car parks was estimated from the government-to-private-EV-charger ratio, which was 34:66 as of October 2019, and the medium-to-quick-charger ratio of non-government car parks, leading to an estimated total budget of HK$130,253,000[8]. The deployment budget was further distributed by partitioning existing charging points across building types. Appendix 1 provides the number of charging stations and the budget allocation for each land-use type considered by this study. The fixed and unit costs of chargers were estimated based on information provided by one of the leading EV charging facility suppliers in Hong Kong during interviews scheduled by the research team.

To obtain an optimal solution, the LA model was processed using IBM ILOG CPLEX. We assessed the solution's performance using the following metrics:.

- **Daily shortfall** (as number of EVs): charging demand that cannot be satisfied by charging facilities (i.e. $\sum_{i\in I} s_i$)
- **Coverage (%)**: percentage of charging demand that can be satisfied by charging facilities (i.e. $\sum_{i\in I}(h_i - s_i)/\sum_{i\in I} h_i$)

4. Results

4.1. Electric vehicle demand analysis

After removing records with missing or invalid data, our final sample included 956 respondents. We used the 'gologit2' STATA package (Williams, 2006) for estimation. Table 2 reports estimates from our generalised ordered probit model following the parsimonious form of the parameterisation proposed by Peterson and Harrell (1990); a full set of estimated Betas for each ordinal category has been estimated but not reported here for simplicity reason. The table's first section presents the Betas (β), which are coefficients of all independent variables. The second section presents the Gammas (γ), which are coefficients of variables that violated the proportional odds assumption at the 5% level of significance, indicating that they could vary across different outcome categories. The value of γ_j of variable X_n is the difference between β_j and β_1 ofX_n. Note that a Gamma equal to 0 for a variable (and, hence, not shown), implies that the parallel lines assumption holds. Variables for each Gamma category represent how, compared to the lower categories, the specific variables affect the probability of higher categories being chosen (Williams, 2006).

As Table 2 illustrates, most explanatory variables for purchase demand appear in the Beta section (which includes all independent variables) and only very few of them appear in the Gamma section (where the coefficient of the variables cannot hold constant across different outcome categories). The estimation output suggests that most of our independent variables satisfied the proportional odds assumption. The two variables (being aged 45 or older and being self-employed) that appear in the Gamma section suggest that the coefficient estimates of these two variables violated the proportional odds assumption, justifying the choice to use the generalised

[7] Government car parks featured 16.48 chargers per station at the time of reference. Thus, it was assumed that fixed costs for 45 car parks would be required to enable the installation of 739 new chargers.

[8] Considering the government–private-charger ratio and the medium-to-quick-charger ratio of non-government car parks, we budgeted for 746 medium chargers and 689 quick chargers in non-government car parks and assigned 249 fixed cost units based on an average of 5.77 chargers per non-government station.

Table 2
Estimates of factors affecting EV purchase demand.

	Coefficients	Std. Err.	z	p-value > |z|
Independent variables				
Beta				
Gender (ref. = male)				
Female	-0.099	0.072	−1.38	0.169
Age (ref. = 18–34)				
Aged 35–44	-0.146	0.091	−1.61	0.107
Aged 45 or above	-0.523***	0.162	−3.22	0.001
Education (ref. = diploma, high school or below)				
Bachelor	0.127	0.082	1.56	0.119
Postgraduate	-0.106	0.107	−0.99	0.323
Marital status (ref. = single and other statuses)				
Married	0.293***	0.086	3.39	0.001
Employment status (ref. = employee/other statuses)				
Self-employed	-0.235	0.233	−1.01	0.313
Housing type (ref. = public housing and other types)				
Private housing	-0.190	0.131	−1.45	0.148
Tenure type (ref. = renters and other types)				
Homeowners	0.245**	0.121	2.03	0.042
Interaction term				
Private housing × Homeowners	0.012	0.156	0.08	0.937
Family size	0.025	0.036	0.70	0.485
Number of children in household	0.073	0.059	1.23	0.219
Income (ref. < HK$40,000)				
HK$40,000 or above	0.116	0.085	1.37	0.170
Residential district characteristics				
Age (average)	-0.147**	0.058	−2.55	0.011
Postsecondary school education (percentage)	0.010	0.007	1.41	0.158
Single (percentage)	0.077**	0.036	2.12	0.034
Hong Kong Island	-0.222*	0.130	−1.70	0.089
New Territories (suburbs)	-0.106	0.105	−1.01	0.315
Workplace district characteristics				
Age (average)	0.046	0.067	0.68	0.494
Postsecondary school education (percentage)	0.004	0.008	0.47	0.639
Single (percentage)	0.009	0.040	0.22	0.824
Hong Kong Island	-0.046	0.138	−0.33	0.739
New Territories (suburbs)	0.035	0.105	0.34	0.736
Gamma				
Gamma 2				
Aged 45 or above	0.113	0.124	0.91	0.361
Self-employed	0.235	0.177	1.32	0.185
Gamma 3				
Aged 45 or above	0.207	0.142	1.45	0.146
Self-employed	0.337	0.211	1.60	0.111
Gamma 4				
Aged 45 or above	0.239	0.156	1.53	0.125
Self-employed	0.404*	0.237	1.71	0.088
Gamma 5				
Aged 45 or above	0.462***	0.165	2.81	0.005
Self-employed	0.630**	0.248	2.54	0.011
Gamma 6				
Aged 45 or above	0.494***	0.183	2.70	0.007
Self-employed	0.889***	0.264	3.37	0.001
Alpha				
cons_1	3.035	3.847	0.79	0.430
cons_2	2.434	3.846	0.63	0.527
cons_3	1.927	3.845	0.50	0.616
cons_4	1.257	3.846	0.33	0.744
cons_5	0.532	3.846	0.14	0.890
cons_6	-0.209	3.846	−0.05	0.957

Note: ***: $p \leq 0.01$; **: $p \leq 0.05$; *: $p \leq 0.1$.
Model summary: N = 956, Pseudo R^2 = 0.0335, Log likelihood = -1694.4751, LR chi^2 = 117.60, Prob > chi^2 = 0.0000.

ordered probit model rather than the ordered probit model. The negative Beta coefficient for the group of older respondents (aged 45 and older) suggests that they were less likely to purchase an EV compared to the younger age group (reference group is aged 18 to 34). This find supports previous observations that younger people favour EVs more than older people (Curtin et al., 2009; Hidrue et al., 2011; Ziegler, 2012). However, for this variable, its γ_5 (0.462) and γ_6 (0.494) are statistically significant at the 5% level (and from another estimate output from 'gologit2′, we know that β_5 (-0.0611) and β_6 (-0.0292) of this variable are both are statistically insignificant). It suggests that respondents aged 45 or above are not less likely to choose "6″ or "7" in the Likert Scale compared to those aged 35 or younger. This additional finding provides more nuanced insight into the age group differences.

As for the Beta coefficient for being self-employed, it was statistically insignificant at the baseline. However, for this variable, its γ_5 (0.630) and γ_6 (0.889) are statistically significant at the 5% level (and from another estimate output from 'gologit2′, we know that β_5 (0.3952) and β_6 (0.6541) of this variable are both are statistically significant). It suggests that self-employed individuals are considerably more likely to express a definite (Likert scale: 7) or very high (Likert scale: 6) inclination to purchase an EV in the near future than employees and those with other employment statuses.

Among variables not violating the proportional odds assumption, several individual and household attributes were statistically significant. For example, compared to single people, married respondents were more likely to buy an EV in the near future. This finding is consistent with car purchase decisions in general in Hong Kong, where car ownership tends to change at significant moments in an individual's life. Meanwhile, compared to tenants of other tenure types, homeowners were generally more likely to buy an EV within the next 5 years. This reflects the priority Chinese society places on homeownership, which means that car ownership only becomes a major ambition upon securing homeownership (e.g. He & Thøgersen, 2017). Regarding the model's district-level variables, although workplace district variables did not produce significant effects, there were several significant residential district variables: people living in a district featuring a younger population or a higher percentage of single individuals were more likely to buy an EV, and respondents residing on Hong Kong Island were less likely to purchase an EV than those living in Kowloon.

4.2. Electric vehicle charging demand

Using the coefficients estimated from the generalised ordered probit model, we forecast aggregate demand at the LSBG level using the sample enumeration procedure based on Eq. (2) with the 2016 micro-census data. We excluded outlying islands from our study area (except for Lantau Island) for further analyses because they do not have a major local economy (except for tourism) and are only connected to Hong Kong's main regions by sea transport. Additionally, where over half of an LSBG's land area was inside the boundaries of country parks, the LSBG would later be excluded from LA modelling due to lack of accessibility[9]. Based on Eq. (3), the total daily public charging demand was projected to be 29,913 EVs. Fig. 4 presents daily public charging demand at the LSBG level, ranging from 2 to 230 EVs. Despite being the wealthiest region, most parts of Hong Kong Island (excluding the Eastern District) indicated only moderate EV charging demand. This can be attributed to public transport providing relatively easy access to workplaces and car parking being expensive; these factors reduce both the need and desire to buy a car in general. In Kowloon, there is substantial charging demand for the East Kowloon area, including Kwun Tong District and parts of Kowloon City District. In the New Territories, new towns represent major sources of charging demand, especially Tseung Kwan O, Tsuen Wan, Tuen Mun, Tin Shui Wai, Tai Po and Tung Chung. A small number of individual residential zones outside the urban areas and new towns (e.g. Discovery Bay) also represent substantial charging demand.

4.3. Results of location-allocation model

We performed our computational experiments on an Intel(R) Core(TM) i7-7700 CPU @ 3.60 GHz personal computer with 32.0 GB of RAM. We adopted CPLEX 12.7.1 as our mixed-integer linear programming solver. Default CPLEX settings were used to solve the LA model with one exception: the termination criterion of an optimality gap of 0.5% was adopted to ensure high solution quality.

We first report the results from the baseline LA model, which CPLEX solved in 164.25 s. Fig. 5 shows the optimal locations of EV charging stations and demand allocation at the LSBG level; Figs. 6 and 7 describe the optimal solution's allocation of new chargers according to charger speed.

According to the optimal solution, in addition to the 332 existing charging stations, 113 new stations would be required to satisfy demand. Of the existing charging stations, 244 would require expansion, indicating accessible locations not reaching supply potential. A total of 1,500 new medium chargers and 698 new quick chargers would be needed at either existing and expanded or new EV charging stations. Current budget and demand projections lead our solution to suggest a total of 4,701 (existing and new) public chargers, a figure resembling the Hong Kong government's target of 5,000 public chargers by 2025 (Hong Kong Environment Bureau, 2021).

Analysis of the performance of the optimal EV charging facility plan indicated a 79.71% demand coverage rate, leaving the daily charging needs of 6,069 EVs unsatisfied. Meanwhile, the model indicated an average travel distance to an EV charging station of 529.82 m.

The spatial characteristics of regions in the territory correspond to varying charger deployment patterns. Most new charging stations recommended by our model would be allocated to the suburban areas to expand charging accessibility in new towns such as Tin Shui Wai, Tuen Mun and Tseung Kwan O. Clusters of new charging stations would also be located in selected communities across the Kowloon and Hong Kong Island urban areas, namely, the north-eastern skirt of Kwun Tong District, Kowloon City District (Kowloon) and Aberdeen (Hong Kong Island). The concentration of new charging stations in these districts suggests that a shortage of convenient charging spots demands new charging locations to promote access to the charging network.

Some areas demonstrate an agglomeration of expanded charging stations. On Hong Kong Island, the Central-Wan Chai CBD

[9] There are 33 of these LSBGs, accounting for 1.48% of the total estimated demand.

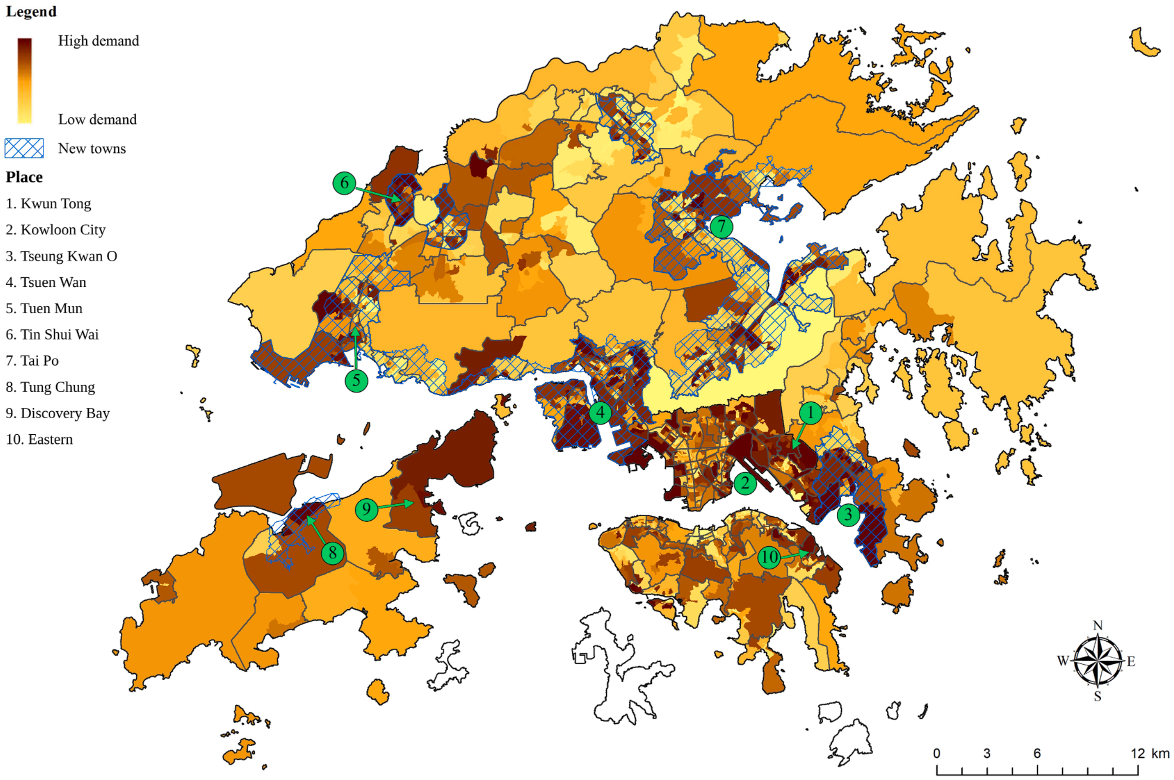

Fig. 4. Map of public charging facility demand at the large street block group level.

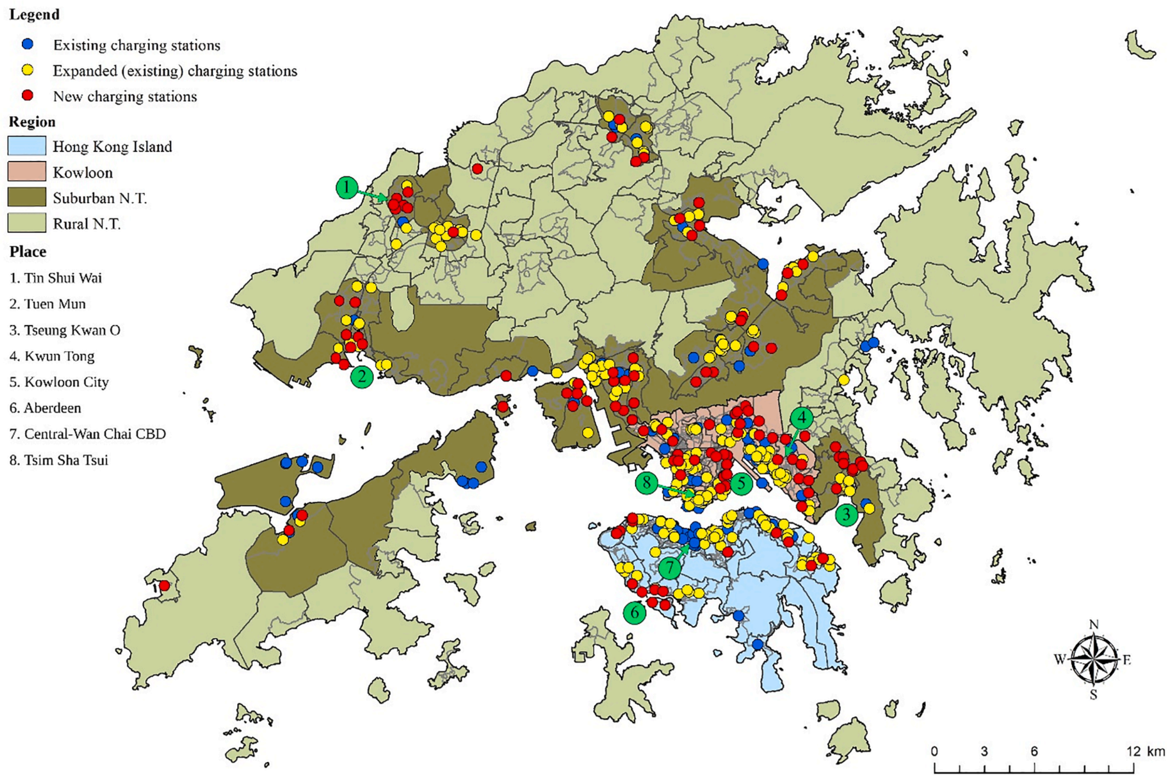

Fig. 5. Distribution of new and existing charging stations in the baseline scenario.

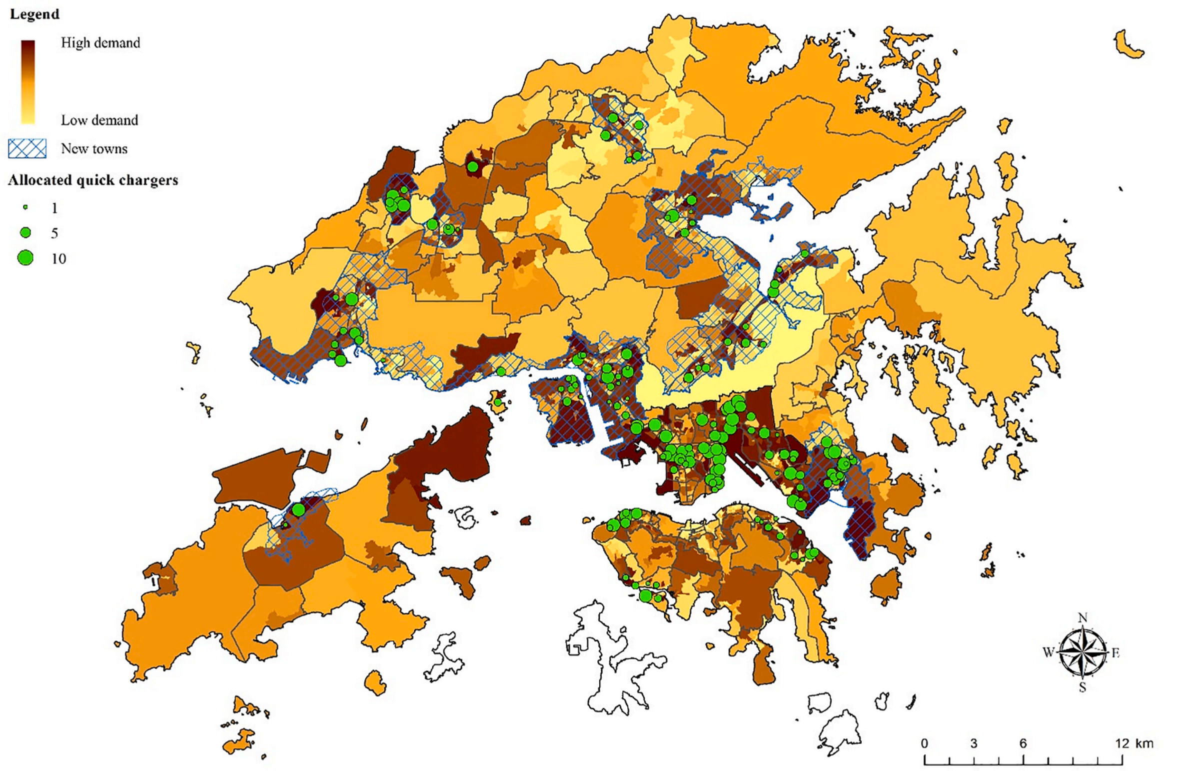

Fig. 6. Distribution of new quick chargers in the baseline scenario.

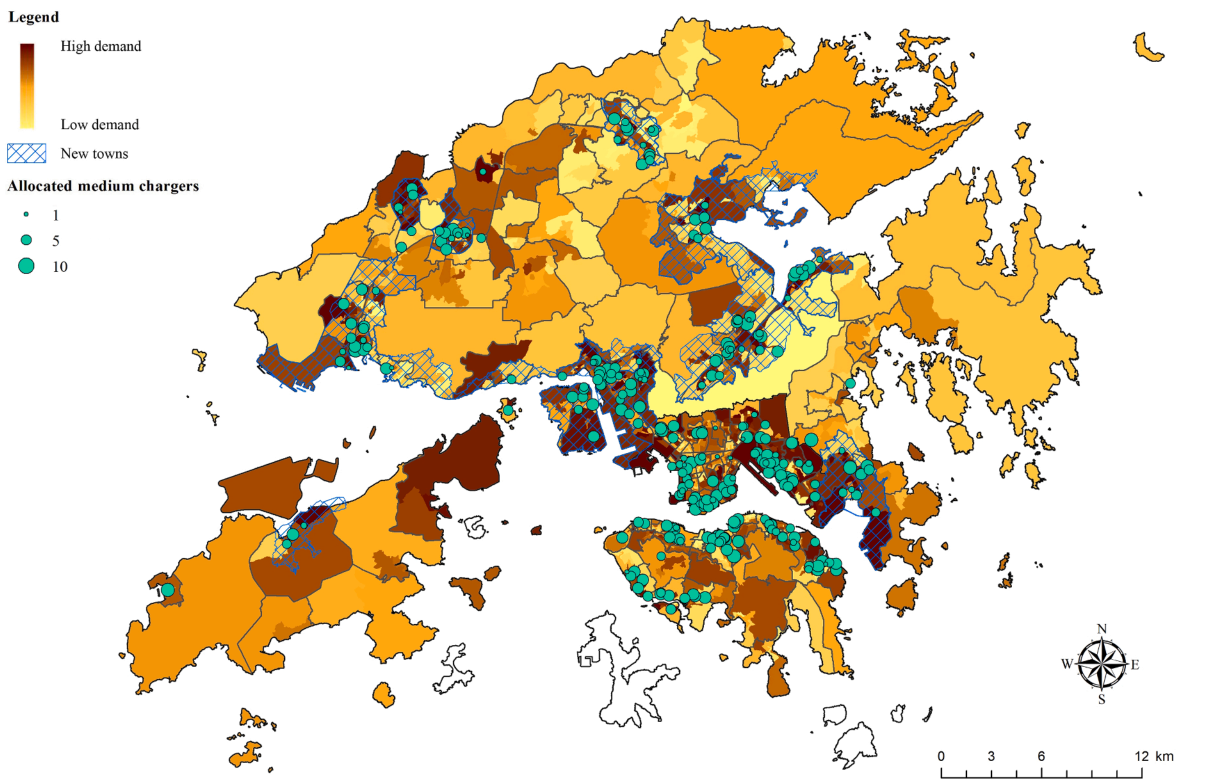

Fig. 7. Distribution of new medium chargers in the baseline scenario.

(Central & Western District and Wan Chai District) would primarily receive charging station expansions, with part of the existing network retained in the optimal plot without any expansion. Two Kowloon districts represent focal points of charging station expansions: Tsim Sha Tsui in the Yau Tsim Mong District and the coastal part of Kwun Tong District. The former is the city's core tourist hub, and the latter is the in-development second CBD. This pattern suggests that these areas enjoy favourable charging locations within the existing network, making it more economical to expand capacity at these stations because it would reduce the fixed costs associated with establishing new charging stations. There could be two possible reasons for the clustering of charging stations expansions: first, these districts enjoy a high concentration of major shopping malls and commercial buildings, which constitute a prominent share of Hong Kong's charging locations (Appendix 1); second, given our model estimated charging demand based on a district's population demographics, business districts, such as the CBD, contain relatively few residential units and its residents do not generate a particularly substantial charging demand.

Analysing charger allocation by type, Figs. 6 and 7 represent the distribution of new quick and medium chargers required to realise the optimisation plan. Of the 357 sites receiving new charger(s), quick chargers would be installed at 158 sites, and medium chargers would be installed at 299 sites. Regarding location allocation, the distribution of quick chargers would closely follow that of the newly established charging stations in Fig. 5, while medium chargers would be distributed more evenly with slightly higher density in urban areas. Comparing the distributions of medium and quick chargers indicates the different roles of the two charger types: quick chargers fill the provision gaps in areas featuring low charging accessibility; medium chargers complement charging capacity in demand hotspots.

We have also examined the need for expanding the EV charging facilities to satisfy future EV charging demand. We ran a case that utilised only existing EV charging infrastructure (i.e. charging stations and piles) to assess current accessibility. This involved setting the budget for each land use type B_l to zero and removing Constraints (6) and (7) from the LA model. Utilising only the existing EV charging infrastructure, the coverage rate, unfulfilled daily charging needs, and average travel distance to an EV charging facility are 27.33%, 21,736 EVs and 619.98 m, respectively. This suggests that the current supply of public charging facilities is insufficient, making it imperative to establish new charging infrastructure – including facilities at both existing and new locations – to satisfy Hong Kong's future EV charging demand. Thus, the budget allocated to establishing EV charging facilities contributes to a 191.66% coverage rate increase, a 72.08% reduction of unsatisfied daily charging demands, and a 14.54% reduction of average travel distance to EV charging stations.

Finally, to examine the LA model's practicality, we compare it with the following approach, which mimics an allocation strategy that policymakers could adopt. This approach first fully allocates existing charging stations to demand points and expands their charging piles, if possible, to satisfy unmet demand. After the number of charging piles has reached the capacity of the charging station, the approach builds new charging stations based on the remaining demand. Such an approach is equivalent to replacing the inequality sign of (4) with an equality sign for all existing charging stations, i.e.,$\forall j \in \overline{J}$. Consequently, this approach produced a coverage rate of 79.00%, a shortage of 6,282 EVs in terms of satisfied daily charging demand, and an average travel distance to an EV charging station of 536.62 m. Although these results are consistent with those of the LA model, the LA model improved upon each metric to provide solutions with better EV charging facility accessibility. Finally, it should be noted that an optimisation approach offers more advantages when the problem is tightly constrained; thus, we expect the model to produce more significant benefits if the authority has a smaller budget for EV charging facility deployment.

5. Sensitivity analysis

To examine the effects of EV charging facility supply and demand on the optimal deployment of chargers, we conducted a sensitivity analysis by varying the budget, demands and coverage radius input into the LA model. These factors have been demonstrated to determine model performance. For example, Dong et al. (2014) repeated model scenarios with varying budget constraints, identifying an ascending deployment share of high-power chargers under expanding budgets, Andrenacci et al. (2016) observed a rescaling effect on model results caused by adjusting EV penetration expectations, and Vazifeh et al. (2019) adjusted the coverage range of charging stations to observe its interrelationship with deployment quantity. For the following sensitivity analysis, all problem settings except for those being tested maintained the same values as the baseline scenario presented in **Section 3.3**. To assess the effects, we measured (i) the number of new stations and chargers of all types and each type (reflecting the impact on the establishment of new charging facilities) and (ii) coverage, daily demand shortfall and average driving distance (reflecting the impact on charging facility accessibility).

5.1. Budget variation

The budget was varied between −50% and +50% at 10% intervals, with Fig. 8 demonstrating the impact of the change on the numbers of new stations and chargers. As expected, budget changes positively impacted the deployment scale of EV charging facilities. The percentage change in budget generally exhibited a linear effect on the number of stations and chargers; however, beyond +40%, the increase in the number of new stations was higher, while the rate of increase in the number of chargers was less significant. This could be attributed to (i) the number of chargers potentially reaching a charging station's capacity, such that more stations were required to increase supply when a higher budget was allocated, and (ii) EV charging demands being almost fulfilled (i.e. the primary objective) when budget was increased by 40%, such that the secondary objective (i.e. travel time to an EV charging facility) directed the solution to establish more charging stations to decrease the average travel time. Given the medium-to-quick-charger ratio was

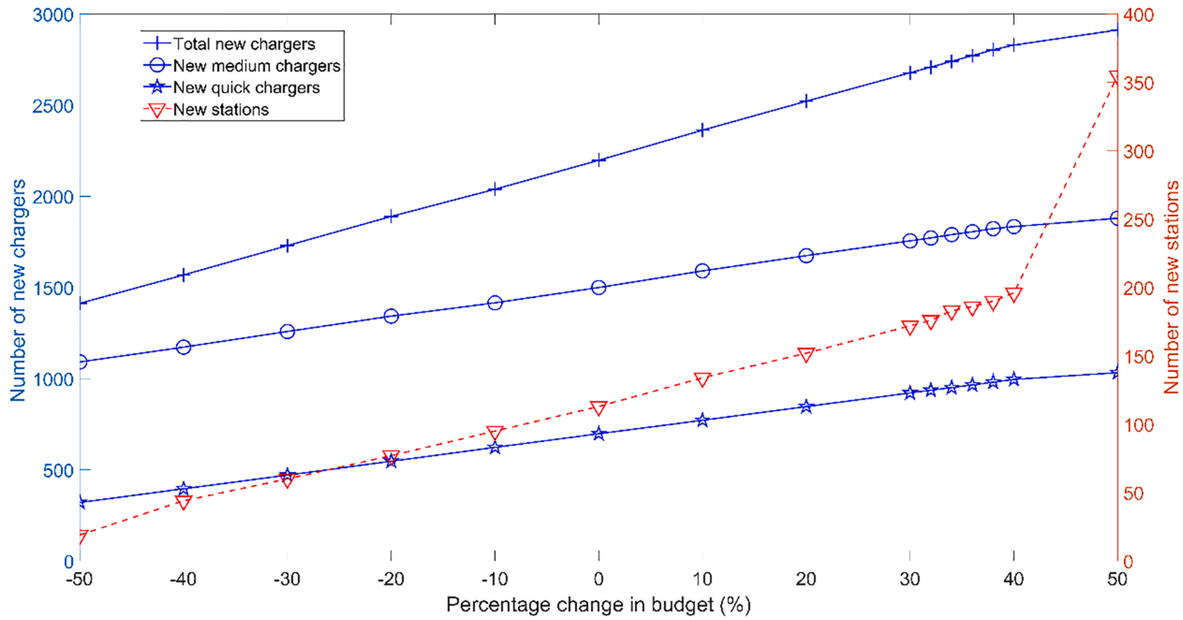

Fig. 8. Impact of budget changes on the number of new stations and chargers.

maintained at 52:48 after allocating the scheduled 739 government-provided chargers, the numbers of medium and quick chargers exhibited similar trends as the percentage budget change increased. In extreme scenarios, when the budget was increased (reduced) by 50%, 355 (19) new stations were established, with 1,880 (1,092) new medium chargers and 1,033 (321) new quick chargers installed. This verifies budget as a critical factor in the expansion of public EV charging networks.

Meanwhile, Fig. 9 represents the impact of budget changes on EV charging demand shortfall and coverage. Unsurprisingly, shortfall decreased and coverage increased alongside budget increases. The effect also appeared to be linear, except that such changes were mild when the budget was increased by over 40% because charging demand coverage almost reached saturation (99.40%). Testing solutions using smaller budget intervals between +30% and +40%, we located an elbow point at the +40% budget level. This elbow point suggests that, in practice, any budget increase beyond 40% of the modelled budget yields minimal improvement to network coverage. Interrogating the impact of varying budget on the average driving distance from a demand point to its allocated charging station revealed only mild changes (between 579.81 m and 663.57 m) following a 50% budget increase or decrease (see Appendix 3). A small

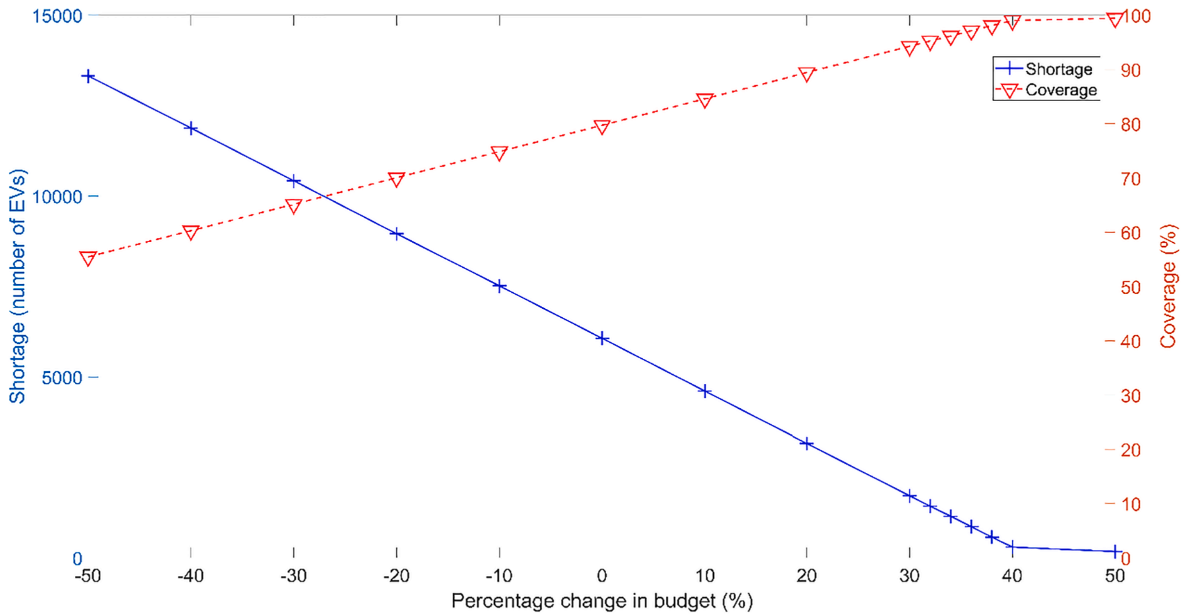

Fig. 9. Impact of budget changes on EV charging demand shortfall and coverage.

but observable change in the trend was identified at the point at which the budget was increased by 40%, a result of driving distances decreasing as more charging stations were established. This slight variation could be explained by the spending priority on expansion over the establishment of new charging stations; that is, the expanded capacity first supplied more remote demand points, and then further budget increases enabled new stations to be constructed to reduce the average driving distance.

5.2. Demand variation

Electric vehicle charging demand was varied from −50% to +50% at 10% intervals; hence, Figs. 10 and 11 are analogous to Figs. 8 and 9, with budget substituted for charging demand.

First, Fig. 10 demonstrates the impact of change in demand on the number of new stations and chargers. As demand increased, the numbers of new chargers (of all types and each type) to be installed increased slightly, and the number of new stations decreased significantly. In the extreme scenarios, when demand was increased (reduced) by 50%, 107 (473) new stations were established, with 1,511 (1,403) new medium chargers and 699 (601) new quick chargers being installed. This illustrates the trade-off between installing new chargers and establishing new charging stations. Assuming a fixed budget, it is more economical to expand existing charging stations to quickly meet demand because installing a medium charger (HK$16,000) costs significantly less than establishing a new charging station (HK$43,500).

Combining this interpretation with the demand satisfaction and coverage plot (Fig. 11) provides a more complete picture. Demand shortfall and coverage demonstrated greater sensitivity to proportional adjustments of demand level than budget. For example, when EV charging demand increased from −50% to +50%, demand shortfall increased from 88 EVs to 20,600 EVs. In contrast, demand coverage decreased from 99.42% to 54.09%. Given the generally constant numbers of new chargers, these statistics indicate that excessive charging demand focuses charger allocation on higher-demand communities, sacrificing coverage because budget constraints already precluded satisfaction of increased charging needs. The notable range of average driving distance – from 398.12 m to 722.18 m (see Appendix 4) – further verifies this interpretation of the impact of varying demand on EV charging facility accessibility. Upon increasing the percentage change in demand from −50% to −20%, the number of new charging stations decreased, meaning charging facilities were located further from demand points. In contrast, when the percentage change in demand increased from −20% to +50%, the number of new charging stations tended to be stable as the percentage change in demand increased. Therefore, the second objective minimised the travel time, reducing the average driving distance to charging facilities.

5.3. Coverage radius variation

The coverage radius of charging stations was varied from −50% to +50% at 10% intervals; hence, Figs. 12 and 13 are analogous to Figs. 8–11, with the percentage change in coverage radius considered in this setting.

Fig. 12 illustrates the impact of the change in coverage radius on the number of new stations and chargers. As coverage radius increased, EV charging stations could cover more of the city, requiring fewer stations and enabling more budget to be allocated to establishing chargers. When demand was increased (reduced) by 50%, 107 (152) new stations were established, with 1,511 (1,498) new medium chargers and 699 (687) new quick chargers being installed. Given the budget was kept constant in this scenario, relatively

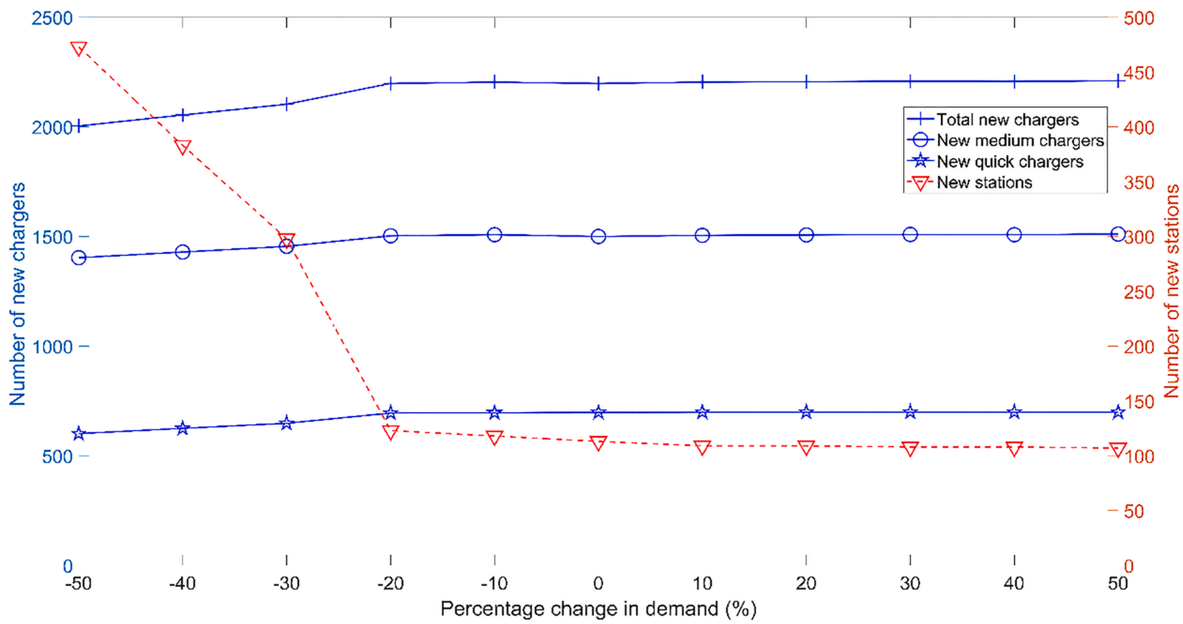

Fig. 10. Impact of changes in demand on the numbers of new stations and chargers.

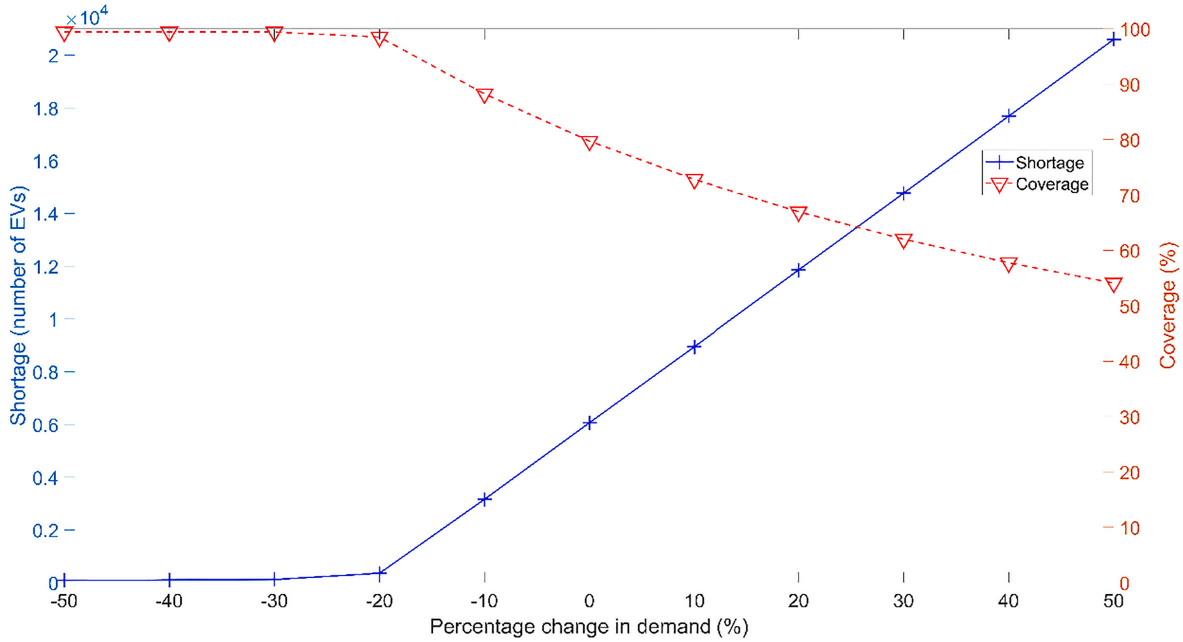

Fig. 11. Impact of changes in demand on EV charging demand shortfall and coverage.

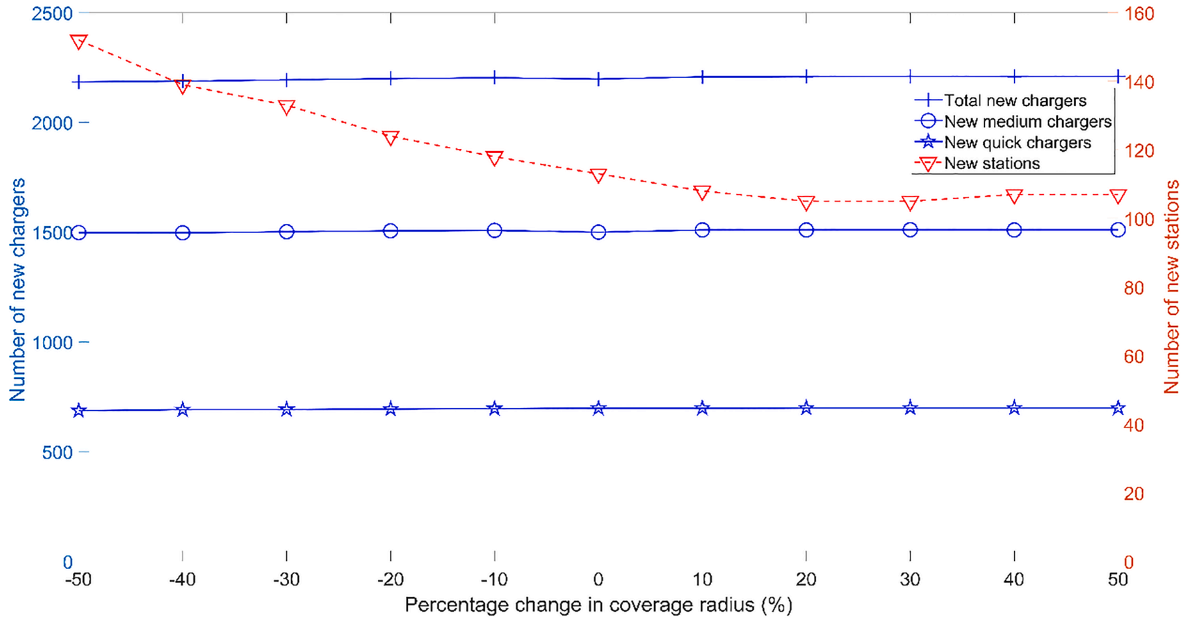

Fig. 12. Impact of changes in coverage radius on the numbers of new stations and chargers.

speaking, increases in the numbers of chargers were not very significant.

Meanwhile, Fig. 13 demonstrates that increases in coverage radius allowed more charging stations to be accessed by EV drivers, reducing unsatisfied charging demand. As expected, the decrease in demand shortfall contributed to greater coverage. For example, when coverage radius was increased from −50% to +50%, demand shortfall reduced from 8,281 EVs to 5,798 EVs, and demand coverage increased from 72.32% to 80.62%. We also considered the impact of coverage radius changes on the average driving distance between demand points and allocated charging stations (see Appendix 5), observing the average driving distance to increase as coverage radius increased. This is because, as coverage radius was increased, EV charging stations further from demand points were recognised as accessible, enabling allocation and increasing the average driving distance.

Based on Figs. 12 and 13 and Appendix 5, it is apparent that changes to coverage radius have a less significant impact – in both the absolute and relative senses – than the changes to budget and EV demand reported in **Sections 5.1 and 5.2**.

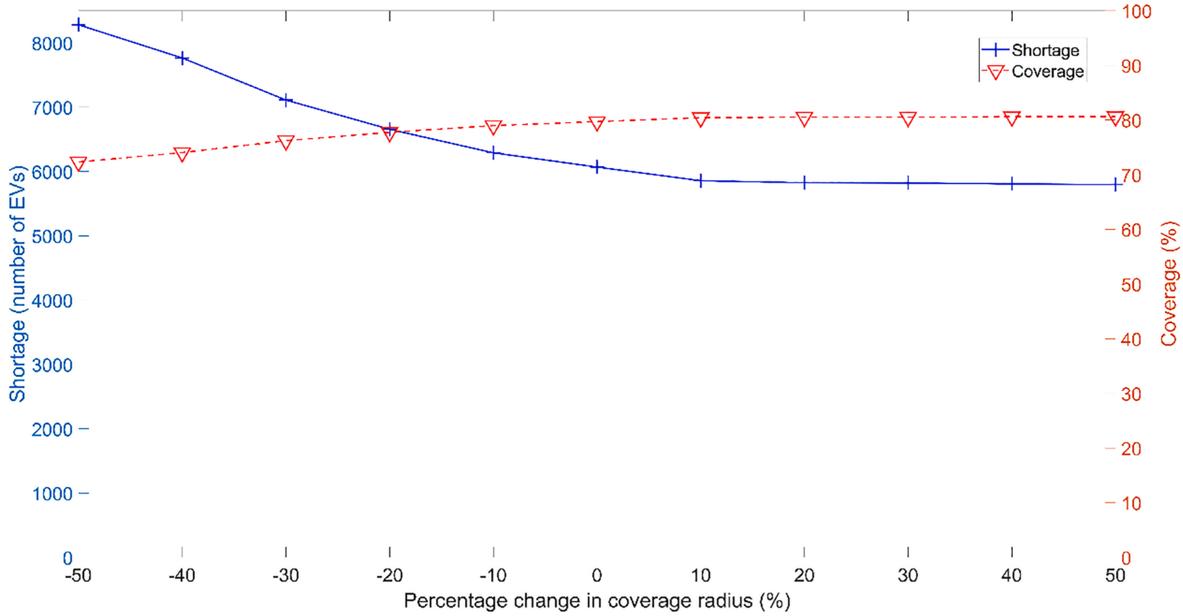

Fig. 13. Impact of changes in coverage radius on EV charging demand shortfall and coverage.

6. Conclusion and policy recommendations

The popularisation of e-mobility (e.g., buses, private vehicles, e-scooters) significantly relies an efficient charging network (e.g., He et al., 2016; Kong et al., 2017; Napoli et al., 2020; Zhang et al., 2021). Focusing on private vehicles, this paper applies a contextualised LA model and empirical solution to the real setting of a high-density city, namely, Hong Kong, which faces a considerable challenge in terms of meeting future EV charging infrastructure demand due to the lack of personal car park and privately owned chargers. Our research presents an interesting case in which public EV charging facilities are essential to popularising e-mobility.

To counter limited home-charger penetration, it is necessary to balance the proximity of public charging facilities to residential areas. In most European and North American cases, public charging stations have mainly been located at park-and-ride hubs, add-on service venues and highway exits (Ai et al., 2018; Kong et al., 2017; Morrissey et al., 2016). However, this approach may not be completely replicable in Asian cities, where public charging networks are expected to fulfil a substantial proportion of the home charging needs of urban dwellers who do not have access to (privately owned) home chargers. This requires charging stations to be preferentially located near residential estates to maximise the feasibility of overnight recharge. Given the interlocking land-use pattern of dense city settings, assigning public EV chargers at community hubs such as shopping malls or community centres could plausibly serve nearby residents and provide add-on services such as quick daytime charging. This spatial deployment recommendation would be relatively easily to adopt in our study area because significant numbers of these locations are managed directly by the government, currently Hong Kong's biggest charging facility provider. To ascertain supply capacity, authorities could assign a minimum percentage of charging-enabled parking spaces to certain communal buildings. For example, Hong Kong's latest planning guideline advise that 30% of private car parking spaces in newly constructed residential, industrial and business developments should be equipped with standard charging facilities (Hong Kong Planning Department, 2021).

Our LA model's results suggest that, based on estimated future public EV charger demand, Hong Kong's existing network should be expanded by 88% (adding 2,198 new chargers to the 2,503 existing ones) to meet the optimal deployment layout. In terms of the spatial distribution of new chargers, we identified several urban districts and new town areas as priority zones. First, the main urban areas of Kowloon and the northern shore of Hong Kong Island are highly accessible by public transport (S. He et al., 2018). Therefore, we recommend the co-location of public EV chargers and park-and-ride infrastructure to encourage better integration of EV with the urban core's existing public transport network to enable a more sustainable and integrated transport network; this could relieve traffic congestion in the central urban areas. Second, fulfilling the demand shortfall in new towns should be prioritised because (i) new towns are projected to represent the major demand hotspots in the next 5 years, which could soon overwhelm existing charging facilities, and (ii) long-distance trips across Hong Kong, especially featuring trips ending in suburban or rural areas, will call for more public charging infrastructure in new towns and new development areas to ease the range anxiety experienced by EV users. Our model aligns with this implication by allocating the majority of new charging stations to new towns and new development areas.

Additionally, our LA model demonstrated that scaling up existing charging stations by incorporating more chargers appears to be more efficient than constructing new charging stations, with our sensitivity analyses confirming that gradually reducing the number of new charging stations would not severely impair facility accessibility, as measured by the driving distance between demand points and allocated charging facilities. That is, the high-density context modelled can maintain facility accessibility under a decreasing number of facilities. This is explained by the ease of charging in surrounding neighbourhoods – our baseline scenario featured an average distance required to reach a charger of fewer than 530 m – given the proximity of developments in compact cities. Thus, our model inclined towards increasing station capacity rather than dispersing the charging network. Financially, by embracing this planning approach, the fixed costs associated with new charging locations could be avoided, and the budget could be focused on installing more chargers at the same locations. Although this approach might deplete the electricity capacity of buildings, demanding that the government and electricity suppliers work to ensure ample power reserves in preparation for charger installations, initiatives are already in place to address the potentiality: since 2017, the Hong Kong government has granted 100% gross floor area concessions to new developments equipped with EV charging enabling infrastructure (e.g. distribution boards and cablings) in underground car parks (Hong Kong Environmental Protection Department, 2020b). Similar policies could be launched in other high-density cities to increase infrastructure readiness.

This study could be improved in several ways. First, charger supply capacity was computed according to the charge duration of two popular EV models (detailed in footnote 4) and without considering temporary obstructions, such as the over-time occupation of charging points. Simulations could incorporate actual charger utilisation schedules if available. Along this vein, more local driving behaviour and charging stations communications could potentially help us better estimate the charging demand and supply (Lim et al., 2021). Second, our model calculated demand allocation solely according to geographical proximity and charging station capacity. Future research could consider the charging venue preferences of EV drivers compared to other potential factors, such as building type. Third, we included estimated demand in the LA model as an exogeneous variable. Future studies could simulate gradual deployment of chargers and the demand shifts afterwards by modelling supply–demand dynamics and heuristics. Fourth, our survey recorded respondent home and workplace locations at the district level, producing a relatively coarse spatial solution that precluded incorporation of potentially relevant variables. Future studies should endeavour to obtain more refined location information at the data collection stage.

CRediT authorship contribution statement

Sylvia Y. He: Conceptualization, Methodology, Formal analysis, Data curation, Supervision, Writing – original draft, Writing – review & editing, Funding acquisition. **Yong-Hong Kuo:** Conceptualization, Methodology, Formal analysis, Writing – original draft. **Ka Kit Sun:** Formal analysis, Data curation, Visualization, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Appendix 1. EV charging location categories and budget allocated by the location-allocation model.

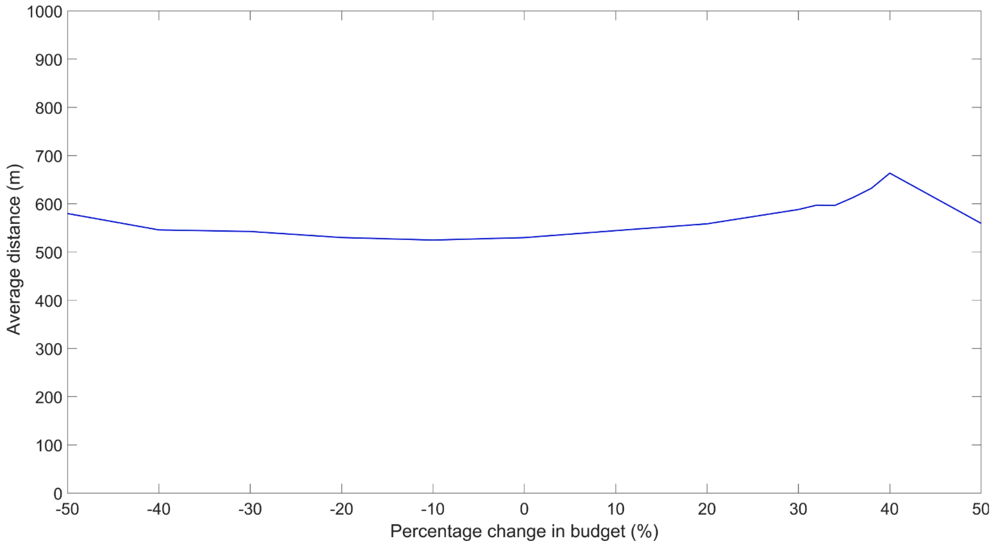
Land use	Location	No. of existing chargers (*ratio*)		Allocated budget (HK$)
Residential	Residential buildings or complexes	524	(*0.209*)	27,222,877
Commercial	Business or commercial centres	912	(*0.364*)	47,412,092
Government and public facilities	Government or institutional buildings	392	(*0.157*)	20,449,721
	Transport hubs	378	(*0.151*)	19,668,203
	Public amenities	19	(*0.008*)	1,042,024
Industrial	Industrial centres	246	(*0.098*)	12,764,794
Others	Comprehensive development areas	32	(*0.013*)	1,693,289

Source: Land-use categorisation based on the Outline Zoning Plan of the Hong Kong Town Planning Board (retrieved from https://www2.ozp.tpb.gov.hk/gos/); locations of charging stations summarised from the October 2019 charger location list published by the Hong Kong Environmental Protection Department.

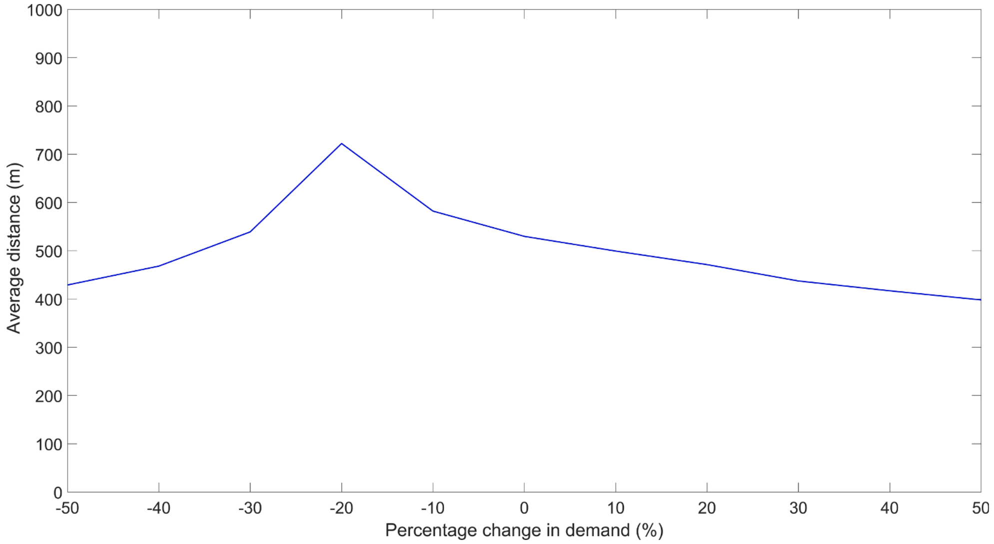
Appendix 2. Personal and household characteristics of survey respondents.

	Percentage (%) or mean			
	EV owners (N = 238)	ICEV owners (N = 327)	Non-car owners (N = 417)	All respondents (N = 982)
Personal characteristics				
Gender				
Male	76.89	57.80	63.79	64.97
Female	23.11	42.20	36.21	35.03
Age				
18 to 34	35.04	42.20	33.17	36.64
35 to 54	55.13	47.71	57.93	53.84
55 and above	9.83	10.09	8.89	9.52
Educational attainment				
Below university	23.11	29.05	36.69	30.86
Bachelor's degree	50.42	51.07	46.28	48.88
Postgraduate degree	26.47	19.88	17.03	20.26
Marital status				
Single	36.60	42.20	37.26	38.75
Married	63.40	57.80	62.74	61.25
Employment status				
Employed	82.28	90.52	87.98	87.45
Self-employed	14.77	7.34	7.21	9.08
Others	2.95	2.14	4.81	3.47
Monthly income (HK$)				
Less than 25,000	21.37	31.69	49.28	36.72
25,000 to 39,999	28.63	35.38	31.25	32.00
40,000 and above	50.00	32.92	19.47	31.28
Household characteristics				
Housing type				
Public rental	11.02	13.76	19.18	15.41
Subsidised home ownership	13.98	16.51	20.62	17.65
Private permanent	75.00	68.20	59.47	66.12
Others	0.00	1.53	0.72	0.82
Housing tenure				
Owner-occupier with an outstanding mortgage	43.64	37.92	39.81	40.10
Owner-occupier with no outstanding mortgage	31.36	34.56	30.70	32.14
Tenant	23.73	25.08	28.54	26.22
Others	1.27	2.45	0.96	1.53
Household size	3.462	3.181	3.139	3.231
Number of children	0.648	0.483	0.499	0.529

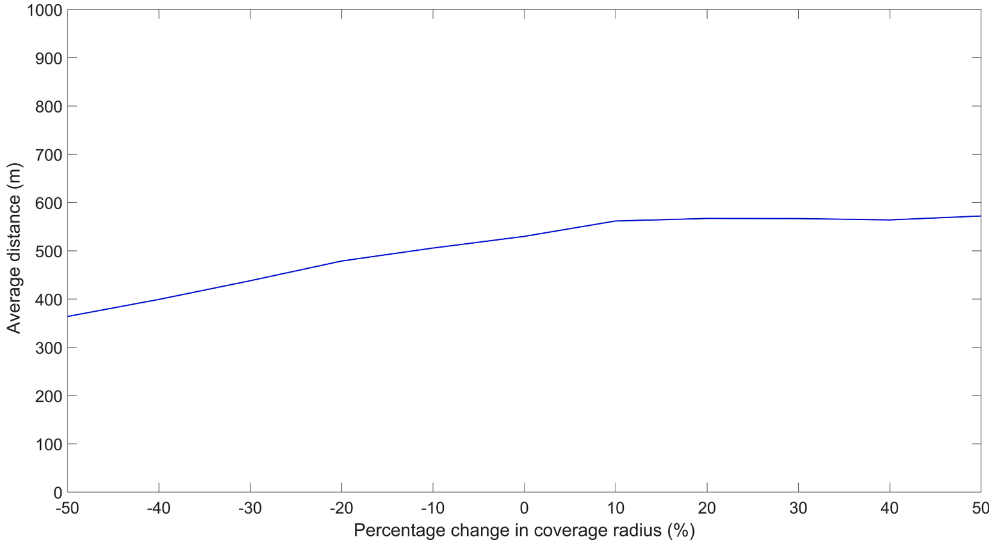
Appendix 3. Impact of budget changes on the average driving distance between demand points and allocated charging facilities.



Appendix 4. Impact of demand changes on the average driving distance between demand points and allocated charging facilities.



Appendix 5. Impact of coverage radius changes on the average driving distance between demand points and allocated charging facilities.



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